

AuthenticEye Forensic Detection Platform

RESEARCH-GRADE MODEL EVALUATION REPORT — GENERATED ON 2026-06-05 16:04:05

Executive Summary

This report documents the performance metrics of the active ensemble classification model running on the AuthenticEye deepfake verification platform. The evaluation was executed on a split validation test suite comprising 40 samples (20 real / 20 synthetic faces). The detector blends features from fine-tuned convolutional neural networks (EfficientNet-B4 & XceptionNet) with signal-level artifacts (FFT frequency ratios & spatial reconstruction statistics) through a fitted Logistic Regression meta-classifier.

Model Configurations & Target Checkpoints

Active Version	v1
Checkpoint Created At	2026-06-05T10:17:14.621118Z
Trained On Dataset	Prepared FF++ & DFDC Split
Hardware Device	CPU Execution (Inference)

Key Classification Metrics

Metric	Value Score
Accuracy	0.4250 (42.50%)
Precision (Positive Predictive Value)	0.4118 (41.18%)
Recall (Sensitivity / TPR)	0.3500 (35.00%)
F1-Score (Harmonic Mean)	0.3784 (37.84%)
Receiver Operating Characteristic (ROC AUC)	0.4862

Performance Visualizations

